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# Analog Electronics Lab

## Laboratory Manual

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|---------------------------------|----------------|
| <b>Course Code</b>              | PCEEL308       |
| <b>Semester</b>                 | S3             |
| <b>Credits</b>                  | 2              |
| <b>Teaching Hours (L:T:P:R)</b> | 0:0:3:0        |
| <b>CIE Marks</b>                | 50             |
| <b>ESE Marks</b>                | 50             |
| <b>Exam Duration</b>            | 2 Hrs. 30 Min. |
| <b>Prerequisite</b>             | Nil            |
| <b>Corequisite</b>              | PBEET304       |

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# Preface

This manual has been prepared to accompany the laboratory course **Analog Electronics Lab (PCEEL308)(2024 scheme)**, offered in Semester 3 of the B.Tech. programme in Electrical and Electronics Engineering. The course is designed to provide hands-on exposure to the design, simulation, and hardware realisation of fundamental analog electronic circuits employing diodes, BJTs, JFETs, MOSFETs, operational amplifiers, and the 555 timer IC.

Each experiment in this manual follows a uniform structure consisting of the aim, theoretical background, pre-lab assignment, components and equipment required, circuit diagram, design procedure, experimental procedure, expected/model graphs, observation tables, results, and viva-voce questions. Students are expected to complete the pre-lab assignment prior to each laboratory session and to record all readings directly in the laboratory record during the conduct of the experiment.

# Course Information

## Course Objectives

1. To design transistor and operational-amplifier based electronic circuits.
2. To simulate and implement the designed circuits in hardware and validate experimental results against theoretical/simulated predictions.

## Course Outcomes

On successful completion of this course, the student will be able to:

| CO  | Course Outcome                                                                | Bloom's Level |
|-----|-------------------------------------------------------------------------------|---------------|
| CO1 | Use the various electronic instruments for conducting experiments.            | K3            |
| CO2 | Design and develop various electronic circuits using diodes and Zener diodes. | K3            |
| CO3 | Design and implement amplifier and oscillator circuits using BJT and JFET.    | K3            |
| CO4 | Design and implement basic circuits using IC (OPAMP and 555 timer).           | K3            |
| CO5 | Simulate electronic circuits using circuit simulation software.               | K3            |
| CO6 | Use PCB layout software for circuit design.                                   | K3            |

Table 1: Course Outcomes and Bloom's Knowledge Level

*K1: Remember, K2: Understand, K3: Apply, K4: Analyse, K5: Evaluate, K6: Create*

## Scheme of Evaluation

### Continuous Internal Evaluation (CIE – 50 Marks)

| Component                                                                                               | Marks     |
|---------------------------------------------------------------------------------------------------------|-----------|
| Attendance                                                                                              | 5         |
| Preparation / Pre-Lab Work, Viva, and Timely Completion of Lab Reports / Record (Continuous Assessment) | 25        |
| Internal Examination                                                                                    | 20        |
| <b>Total</b>                                                                                            | <b>50</b> |

Table 2: CIE Marks Distribution

The continuous assessment component (25 marks) is further broken down as:

- **Preparation and Pre-Lab Work (7 Marks):** Pre-lab assignments/quizzes and understanding of theoretical background.
- **Conduct of Experiments (7 Marks):** Adherence to procedure, accurate execution, safety, skill proficiency, and teamwork.
- **Lab Reports and Record Keeping (6 Marks):** Clarity, completeness, accuracy, and timely submission.
- **Viva Voce (5 Marks):** Ability to explain the experiment, results, and underlying principles.

The final marks for preparation, conduct, viva, and record are computed as the average across all experiments listed in the syllabus.

### End Semester Examination (ESE – 50 Marks)

| Component                                                                 | Marks     |
|---------------------------------------------------------------------------|-----------|
| Procedure / Preliminary Work / Design / Algorithm                         | 10        |
| Conduct of Experiment / Execution of Work / Troubleshooting / Programming | 15        |
| Result with Valid Inference / Quality of Output                           | 10        |
| Viva Voce                                                                 | 10        |
| Record                                                                    | 5         |
| <b>Total</b>                                                              | <b>50</b> |

Table 3: ESE Marks Distribution

- Students shall be permitted to appear for the end-semester examination only upon submission of a duly certified record.
- The external examiner shall endorse the record at the time of the examination.

**CO–PO Mapping**

|     | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|
| CO1 | 3   |     |     |     |     |     |     |     |     |      |      |      |
| CO2 | 2   | 3   | 3   | 3   | 3   |     |     |     | 3   | 3    |      |      |
| CO3 | 2   | 3   | 3   | 3   | 3   |     |     |     | 3   | 3    |      |      |
| CO4 | 2   | 3   | 3   | 3   | 3   |     |     |     | 3   | 3    |      |      |
| CO5 | 2   | 3   | 3   | 3   | 3   |     |     |     | 3   | 3    |      |      |
| CO6 | 3   |     |     |     |     |     |     |     | 3   | 3    |      |      |

Table 4: CO–PO Mapping (1: Slight, 2: Moderate, 3: Substantial, blank: No Correlation)

**Text Books and References**

1. Robert T. Paynter, *Introductory Electronic Devices and Circuits*, Pearson Education.
2. Boylestad R. L. and Nashelsky L., *Electronic Devices and Circuit Theory*, Pearson Education.
3. Donald A. Neamen, *Electronic Circuits: Analysis and Design*, McGraw Hill.

# General Instructions to Students

## Laboratory Conduct

1. Students must bring the laboratory record, observation notebook, graph sheets, and a scientific calculator to every session.
2. The pre-lab assignment for the experiment scheduled for the day must be completed and ready for evaluation before entering the laboratory.
3. Students shall enter the laboratory only with permission and shall occupy the workbench allotted to them.
4. Circuit connections must be checked and approved by the faculty-in-charge or laboratory instructor before the power supply is switched on.
5. Readings must be entered directly into the observation notebook at the time of the experiment; fabricated or estimated readings will not be accepted.
6. Calculations, graphs, and inferences must be completed and the experiment got corrected on the same day, wherever possible, or within the stipulated time announced by the instructor.
7. Students must simulate the circuit using the prescribed circuit simulation software prior to hardware implementation, wherever applicable.
8. Mobile phones shall be kept in silent mode and shall not be used during the conduct of the experiment, except as directed for simulation work.
9. Any damage to equipment or components due to negligence shall be reported immediately and may be subject to the institution's equipment-damage policy.
10. Students are responsible for returning all components, probes, and patch cords to their designated locations and for switching off all instruments and the workbench power supply before leaving the laboratory.

## Best Practices for Breadboarding

Breadboarding is a fundamental skill in analog electronics. A neat and disciplined approach in breadboarding will help achieving the expected results without any hassle. Following these best practices will help you build reliable circuits and simplify troubleshooting:

- **Plan Your Layout:** Before placing components, mentally (or physically with a drawing) plan the layout of your circuit on the breadboard. Group related components.
- **Shortest Connections:** Use short, neat wires for connections. Long, messy wires can introduce noise, capacitance, and inductance, and make troubleshooting difficult.

- **Color Coding:** Use different colored wires for different purposes (e.g., red for +Vcc, black for ground, blue for signal lines). This greatly aids in tracing connections.
- **Power Rails:** Always connect the power supply rails (+Vcc and GND) to the dedicated power buses on the breadboard.
- **Component Orientation:** Pay close attention to the polarity of components like diodes, LEDs, and electrolytic capacitors.
- **Secure Connections:** Ensure all component leads and wires are firmly inserted into the breadboard holes. Loose connections are a common cause of circuit malfunction.
- **Start Simple:** When troubleshooting, simplify your circuit. Isolate sections and test them individually.
- **Component Identification:** Familiarize yourself with common component packages and their pinouts (e.g., op-amp pin configurations, transistor pinouts).
- **Avoid Overcrowding:** Don't try to cram too many components into a small area. Use a larger breadboard if necessary.
- **Check Continuity:** Before applying power, use a Digital Multimeter (DMM) to check for short circuits between power rails and to verify critical connections.

## Observations and Data Recording

Accurate and detailed observations are crucial for a successful lab experience and for writing comprehensive lab reports.

- **Lab Notebook:** Maintain a dedicated lab notebook.
- **Date and Time:** Always start each entry with the date and time.
- **Experiment Title:** Clearly state the experiment title and number.
- **Circuit Diagrams:** Draw clear and labeled circuit diagrams.
- **Component Values:** Record all component values used in your circuit.
- **Measurements:** Tabulate all measured data with appropriate units.
- **Waveforms:** Sketch or capture screenshots of oscilloscope waveforms, labeling axes, amplitudes, and frequencies.
- **Unexpected Results:** Note down any unexpected observations or deviations from expected results. This is often where valuable learning occurs.
- **Troubleshooting Steps:** Document any troubleshooting steps taken and their outcomes.
- **Inferences and Discussions:** Write down your inferences and discuss the implications of your observations.

## Lab Report (Record) Submission

1. The first thing to note is that the lab report/record is **NOT A COPY** of the lab manual. The laboratory record is, as its name indicate, a record/report of the experiment done in the lab. The lab manual may include information required for aiding the writing of the report.
2. The lab record must contain, for each experiment: aim, components/equipment list, design, circuit diagram (with all component labels/values clearly shown), experimental procedure, observation tables, model graphs, sample calculations, result, and inferences. Please note that it is not required to write the theory and informational notes that appear in this lab manual.
3. It is suggested to draw the circuit diagrams and observation tables on the left side of the record, and preferably **not on a graph page**. Graph pages are for drawing graphs, waveforms and sample/model curves/graphs.
4. Records must be certified by the faculty-in-charge within one week of completion of the experiment.
5. Submission of a duly certified record is mandatory for eligibility to appear for the end-semester examination.

## Evaluation of Pre-Lab Assignments

Two pre-lab assignments are prescribed by the syllabus prior to the regular experiments:

1. Measurement of current, voltage, frequency, and phase shift of a signal in an RC network using a cathode-ray oscilloscope (CRO) / digital storage oscilloscope (DSO).
2. Introduction to circuit simulation using a circuit simulation software package (e.g., LTspice, PSpice, Multisim, or NgSpice/QUCS).

Students are expected to complete these foundational exercises before attempting the numbered experiments listed in Table 6.

# Safety Guidelines

Safety is paramount in any laboratory environment. Please read and adhere to the following safety guidelines:

## Safety Note

1. **No Food or Drink:** Do not bring or consume food or drinks in the lab.
2. **Work Area:** Keep your workspace clean and organised. Cluttered benches can lead to accidents.
3. **Before Power On:** Always verify the power supply voltage and polarity before connecting it to the circuit. Reversed polarity can permanently damage diodes, transistors, op-amps, and ICs.
4. **Power Off First:** Switch off the power supply before making, modifying, or removing any circuit connections.
5. **Component Handling:** Handle components carefully. Be aware of the polarity of diodes, LEDs, and electrolytic capacitors. Do not exceed the maximum voltage, current, or power ratings of any active or passive component. Always compute and verify current-limiting resistor values before energising a circuit. Use current-limiting resistors in series with LEDs, diodes, and Zener diodes during testing to prevent excessive current flow. Electrolytic capacitors are polarised. Verify polarity before connecting; reversed polarity may cause the capacitor to overheat, vent, or rupture.
6. **Tool Usage:** Use tools (wire strippers, pliers, screwdrivers) correctly and for their intended purpose.
7. **No Short-circuits:** Avoid short-circuiting the output terminals of a regulated DC power supply or the output of an operational amplifier.
8. **Hot Components:** Be aware of the heat-sink and case temperature of power transistors and voltage regulators during prolonged operation; avoid direct contact.
9. Do not work on live circuits with wet hands or in a wet environment.
10. **Report Accidents:** In the event of a component overheating, emitting smoke, or unusual odour, immediately switch off the power supply and inform the laboratory instructor before attempting further investigation. Immediately report any accidents, spills, or equipment malfunctions to the lab instructor or faculty-in-charge or TA.
11. **No Horseplay:** Maintain a professional demeanour and avoid any behaviour that could endanger yourself or others.

12. **Emergency:** Be aware of the location of emergency exits and first-aid kits. Familiarise yourself with the location of the mains isolation switch and the fire extinguisher in the laboratory.
13. **Before Leaving:** Ensure your workstation is clean and all equipment is turned off before leaving the lab.

#### Precautions

1. Identify the pin configuration (pinout) of every IC and transistor from its datasheet before inserting it into the breadboard.
2. Avoid loose or floating connections, particularly at the input terminals of operational amplifiers, since floating CMOS/op-amp inputs may result in unpredictable circuit behaviour.
3. Keep signal leads and connecting wires as short as practicable to minimise stray capacitance and inductance, particularly in high-frequency oscillator circuits.
4. Re-verify all connections against the circuit diagram before switching on the power supply, and recheck after every modification.
5. Handle CRO/DSO probes with care; ensure the ground (common) lead of the probe is connected to the circuit ground/common point to avoid ground loops and inaccurate measurements.

# List of Equipment and Software

## Introduction to Essential Lab Equipment

Familiarity with the following equipment is essential for performing experiments in this lab.

### Digital Storage Oscilloscope (DSO)

- **Purpose:** The DSO is used to visualize and analyze time-varying electrical signals (waveforms). It displays voltage as a function of time.
- **Key Controls:**
  - **Vertical (Volts/Div):** Controls the voltage scale on the screen (voltage per vertical division).
  - **Horizontal (Sec/Div):** Controls the time scale on the screen (time per horizontal division).
  - **Trigger:** Stabilizes repetitive waveforms on the screen. Adjust trigger level and mode (edge, auto, normal).
  - **Channels:** DSOs typically have multiple input channels (CH1, CH2, etc.) to view multiple signals simultaneously.
  - **Cursors/Measurements:** Use cursors to precisely measure voltage and time intervals. Most DSOs have built-in automated measurement functions (e.g.,  $V_{pp}$ ,  $V_{rms}$ , frequency, period).
- **Usage Tips:**
  - Always use a proper probe (1X or 10X). Understand the attenuation factor. Calibrate probes if necessary.
  - Ground connections are crucial for accurate measurements.
  - Start with auto-set if you're unsure of the signal characteristics, then fine-tune.

### Function Generator

- **Purpose:** Generates various standard waveforms (sine, square, triangle, pulse) at specified frequencies and amplitudes.
- **Key Controls:**
  - **Waveform Selection:** Selects the desired output waveform.
  - **Frequency:** Sets the output frequency.
  - **Amplitude:** Sets the peak-to-peak voltage of the output waveform.
  - **DC Offset:** Adds a DC voltage component to the output signal.
  - **Output Impedance:** Typically 50  $\Omega$ . Be aware of impedance matching.

### Usage Tips:

- Connect the output of the function generator to your circuit's input.
- Always check the output on the oscilloscope before connecting to your circuit to ensure it's set correctly.

## DC Power Supply

- **Purpose:** Provides stable DC voltage and current to power your circuits.
- **Key Controls:**
  - **Voltage Adjustment:** Sets the output voltage.
  - **Current Limit:**<sup>1</sup> Sets/limits the maximum current the supply can deliver. This is a crucial safety feature to protect your circuit from excessive current.
  - **Output On/Off:** Turns the power supply output on or off.

### Usage Tips:

- Always set the voltage and current limit before connecting to your circuit. Start with a lower voltage and gradually increase if needed.
- If the current limit indicator lights up, it means your circuit is drawing more current than allowed, possibly indicating a short circuit or fault. Turn off power immediately and troubleshoot.

## Digital Multimeter (DMM)

- **Purpose:** Measures various electrical parameters like voltage (AC/DC), current (AC/DC), resistance, and continuity.
- **Key Functions:**
  - **Voltage Measurement (V):** Connect in parallel with the component or circuit portion you want to measure. Select AC/DC option appropriately.
  - **Current Measurement (A):** Connect in series with the circuit. This requires breaking the circuit. Be cautious about the current range.
  - **Resistance Measurement ( $\Omega$ ):** Measure resistance of disconnected components. Never measure resistance in an active circuit.
  - **Continuity Test:** Beeps if there is a low-resistance path (useful for checking connections).

### Usage Tips:

- Always select the correct function (V, A,  $\Omega$ ) and range before making a measurement.
- For current measurements, always connect the DMM in series. Incorrect connection can blow the internal fuse.
- Ensure the circuit is de-energized when measuring resistance.

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<sup>1</sup>Note that by adjusting this control, one cannot vary the current. The current drawn from a voltage-source power supply is decided by the load resistance.

| <b>Equipment / Software / Components</b>                                                                     | <b>Typical Use in this Lab</b>                                           |
|--------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| Regulated DC Power Supply (dual/triple output)                                                               | Biasing of transistor, op-amp, and 555 timer circuits                    |
| Function / Signal Generator                                                                                  | Input excitation for amplifiers, filters, and frequency-response studies |
| Digital Storage Oscilloscope (DSO)                                                                           | Measurement of waveforms, amplitude, frequency, and phase shift          |
| Digital Multimeter (DMM)                                                                                     | DC bias point and resistance/voltage/current measurements                |
| Breadboard and connecting wires                                                                              | Circuit prototyping                                                      |
| Active and Passive Components (diodes, Zener diodes, BJTs, JFETs, MOSFETs, resistors, capacitors, inductors) | Circuit construction across experiments                                  |
| Operational Amplifier ICs (e.g., $\mu A741$ , LM358)                                                         | Op-amp based experiments                                                 |
| Comparator IC (e.g., LM311)                                                                                  | Comparator and Schmitt trigger experiments                               |
| 555 Timer IC                                                                                                 | Astable and monostable multivibrator experiments                         |
| Circuit Simulation Software (e.g., LTspice / Qspice / NgSpice-QUCS)                                          | Pre-implementation simulation of all designed circuits                   |
| PCB Layout Software (e.g., KiCad )                                                                           | Introduction to PCB design                                               |

Table 5: Indicative list of equipment and software used in the course

# List of Experiments

Table 6: List of Experiments – Analog Electronics Lab (PCEEL308)

| Expt. No. | Title                                                                                                                   | CO  |
|-----------|-------------------------------------------------------------------------------------------------------------------------|-----|
| PL-1      | Measurement of current, voltage, frequency, and phase shift of a signal in an RC network using a CRO/DSO (Pre-Lab)      | CO1 |
| PL-2      | Introduction to circuit simulation using a circuit simulation software (Pre-Lab)                                        | CO5 |
| 1         | Clipping and clamping circuits using diodes                                                                             | CO2 |
| 2         | Basic RC circuits – High-pass and Low-pass filters                                                                      | CO1 |
| 3         | RC-coupled amplifier using BJT in CE configuration – Measurement of gain, bandwidth, and plotting of frequency response | CO3 |
| 4         | Emitter follower amplifier                                                                                              | CO3 |
| 5         | JFET amplifier – Measurement of gain, bandwidth, and plotting of frequency response                                     | CO3 |
| 6         | MOSFET amplifier                                                                                                        | CO3 |
| 7         | Design and testing of voltage regulators – Zener and series                                                             | CO2 |
| 8         | Design and setup of inverting and non-inverting amplifiers                                                              | CO4 |
| 9         | Op-amp circuits – Scale changer, adder, integrator, and differentiator                                                  | CO4 |
| 10        | Precision rectifier using op-amp                                                                                        | CO4 |
| 11        | Op-amp oscillators – RC phase-shift and Wien bridge oscillator                                                          | CO4 |
| 12        | Op-amp oscillator – LC oscillator (Colpitts or Hartley)                                                                 | CO4 |
| 13        | Waveform generation – Square, triangular, and sawtooth waveform generation using op-amps                                | CO4 |
| 14        | Basic comparator and Schmitt trigger circuits using op-amp (using comparator ICs such as LM311)                         | CO4 |
| 15        | Active filters (one each of high-pass and low-pass)                                                                     | CO4 |
| 16        | Instrumentation amplifier                                                                                               | CO4 |
| 17        | Astable and monostable multivibrator circuits using 555 timer IC                                                        | CO4 |
| 18        | Introduction to PCB layout software                                                                                     | CO6 |

## Format Followed for Each Experiment

Each experiment in the chapters that follow is presented under the following uniform headings:

1. **Aim**
2. **Pre-Lab Assignment**

3. **Components and Equipment Required**
4. **Theory**
5. **Circuit Diagram**
6. **Design**
7. **Procedure**
8. **Precautions**
9. **Model Graph / Expected Waveform**
10. **Observation Table**
11. **Result**
12. **Questions**

The students are expected to submit the report of each experiment, containing the following:

1. **Aim**
2. **Pre-Lab Assignment**
3. **Components and Equipment Required**
4. **Circuit Diagram**
5. **Design**
6. **Procedure**
7. **Model Graph / Expected Waveform**
8. **Observation Tables**
9. **Results/Conclusions**

## Experiment 1

# Astable Multivibrator Using IC 555

### 1.1 Aim

To design, construct, and evaluate an Astable Multivibrator (Free-Running Oscillator) circuit using the IC 555 timer for the given frequency and duty ratio specifications.

1. To observe and record the output square wave and timing capacitor exponential charge/discharge voltage waveforms on a Digital Storage Oscilloscope (DSO).
2. To measure the high time ( $t_1$ ), low time ( $t_2$ ), overall time period ( $T$ ), output frequency ( $f_0$ ), and duty cycle ( $D$ ).
3. To verify upper ( $V_{TH}$ ) and lower ( $V_{TL}$ ) trigger thresholds relative to the supply voltage  $V_{CC}$ .
4. To compare experimentally measured timing parameters with theoretical values derived from component values.

### 1.2 Pre-Lab Assignment

Before conducting the laboratory session, complete the following analytical problems:

1. Sketch the internal block diagram of IC 555 showing the voltage divider network, two comparators, RS flip-flop, discharge transistor, and output driver stage.
2. Derive the expressions for the capacitor charging time ( $t_1$ ), discharging time ( $t_2$ ), total time period ( $T$ ), frequency ( $f_0$ ), and duty cycle ( $D$ ) for an astable multivibrator.
3. Calculate  $R_A$ ,  $R_B$ , and  $C$  required to generate a square wave of  $f_0 = 1$  kHz with a duty cycle  $D = 66.7\%$  using a supply voltage of  $V_{CC} = +5$  V.
4. Explain why a standard 555 astable multivibrator circuit cannot achieve a duty cycle of 50% or less, and describe how a steering diode can be added to achieve a 50% duty cycle.

### 1.3 Components and Equipment Required

The required components and test instruments are listed in Table 1.1.

### 1.4 Theory

An astable multivibrator is a regenerative circuit with two quasi-stable states. It oscillates continuously between these states without requiring external triggering. When constructed using the IC 555 timer, timing is controlled by an external RC network ( $R_A$ ,  $R_B$ , and  $C$ ).

Table 1.1: Components and Equipment Specifications

| S. No. | Item Name                    | Specification / Range          | Quantity    |
|--------|------------------------------|--------------------------------|-------------|
| 1      | Timer IC                     | IC 555 (8-pin DIP)             | 1 No.       |
| 2      | Resistors                    | 4.7 k $\Omega$ (1% Tolerance)  | 2 Nos.      |
| 3      | Timing Capacitor             | 0.1 $\mu$ F (Ceramic/Film)     | 1 No.       |
| 4      | Bypass Capacitor             | 0.01 $\mu$ F (Ceramic)         | 1 No.       |
| 5      | Regulated DC Power Supply    | +5 V or +12 V DC               | 1 No.       |
| 6      | Digital Storage Oscilloscope | Dual Channel, 20 MHz or higher | 1 No.       |
| 7      | Digital Multimeter           | 4 $\frac{1}{2}$ digit display  | 1 No.       |
| 8      | Breadboard & Wires           | Standard setup                 | As required |

### 1.4.1 Principle of Operation

The internal voltage divider of the 555 IC sets reference thresholds of  $V_{TH} = \frac{2}{3}V_{CC}$  for Comparator 1 (Threshold) and  $V_{TL} = \frac{1}{3}V_{CC}$  for Comparator 2 (Trigger).

- **Charging Phase (High Output State):** Initially, capacitor  $C$  is uncharged ( $V_C < \frac{1}{3}V_{CC}$ ). Comparator 2 sets the internal RS flip-flop, forcing Output (Pin 3) HIGH ( $V_{OH} \approx V_{CC}$ ) and turning the internal discharge transistor (Pin 7) OFF. Capacitor  $C$  charges exponentially towards  $V_{CC}$  through resistors  $R_A$  and  $R_B$ . The charging time  $t_1$  required for  $V_C$  to rise from  $\frac{1}{3}V_{CC}$  to  $\frac{2}{3}V_{CC}$  is:

$$t_{ON} = \ln(2) \cdot (R_A + R_B)C \approx 0.693(R_A + R_B)C \quad (1.1)$$

- **Discharging Phase (Low Output State):** When  $V_C$  reaches  $\frac{2}{3}V_{CC}$ , Comparator 1 resets the flip-flop, driving Output (Pin 3) LOW ( $V_{OL} \approx 0$  V) and turning the discharge transistor (Pin 7) ON (connecting it to GND). Capacitor  $C$  discharges exponentially towards 0 V through  $R_B$  into Pin 7. The discharge time  $t_2$  required for  $V_C$  to fall from  $\frac{2}{3}V_{CC}$  to  $\frac{1}{3}V_{CC}$  is:

$$t_{OFF} = \ln(2) \cdot R_B C \approx 0.693 R_B C \quad (1.2)$$

### 1.4.2 Frequency and Duty Cycle Derivations

The total period of oscillation  $T$  is the sum of charging and discharging durations:

$$T = t_{ON} + t_{OFF} = 0.693(R_A + 2R_B)C \quad (1.3)$$

The overall frequency of oscillation  $f$  is given by:

$$f = \frac{1}{T} = \frac{1.44}{(R_A + 2R_B)C} \quad (1.4)$$

The duty cycle  $D$ , defined as the ratio of HIGH output duration to total period, is:

$$D = \frac{t_1}{T} \times 100\% = \frac{R_A + R_B}{R_A + 2R_B} \times 100\% \quad (1.5)$$

Since  $R_A > 0$ ,  $D$  is always strictly greater than 50%<sup>1</sup>.

## 1.5 Circuit Diagram

The schematic for the 555 Astable Multivibrator circuit is presented in Figure 1.1.

<sup>1</sup>There are ways in which duty ratio  $D \leq 50\%$  can be achieved.

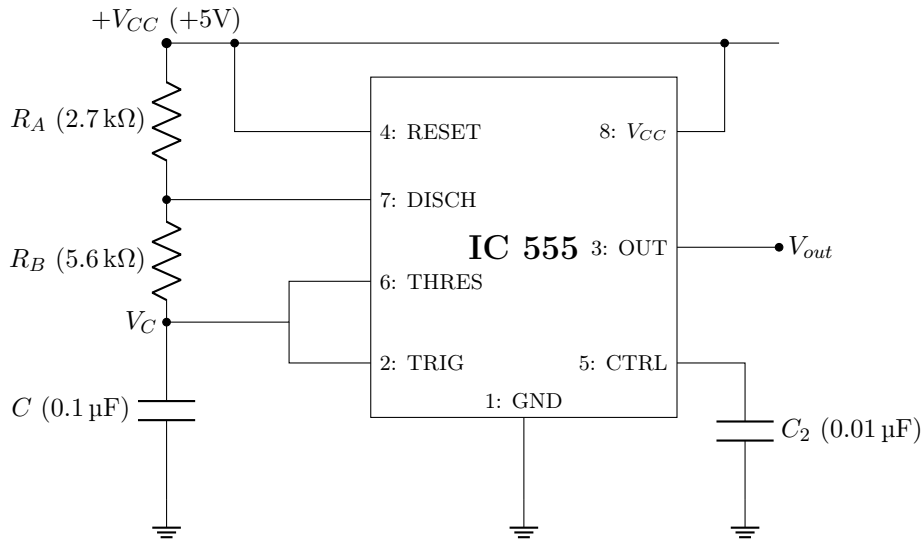


Figure 1.1: IC 555 Astable Multivibrator Schematic

## 1.6 Design

### 1.6.1 Design Specifications

- Target Output Frequency:  $f_0 = 1 \text{ kHz} \implies T = 1.0 \text{ ms}$
- Target Duty Cycle:  $D = 60\% = 0.60$
- Power Supply Voltage:  $V_{CC} = +5 \text{ V DC}$
- Chosen Timing Capacitor:  $C = 0.1 \mu\text{F}$

### 1.6.2 Calculations

#### Timing Equations

- **High Output Duration ( $t_{ON}$ ):** Capacitor charges from  $\frac{1}{3}V_{CC}$  to  $\frac{2}{3}V_{CC}$ :

$$t_{ON} = \ln(2) \cdot (R_A + R_B)C \approx 0.693(R_A + R_B)C \quad (1.6)$$

- **Low Output Duration ( $t_2$ ):** Capacitor discharges from  $\frac{2}{3}V_{CC}$  to  $\frac{1}{3}V_{CC}$ :

$$t_{OFF} = \ln(2) \cdot R_B C \approx 0.693 R_B C \quad (1.7)$$

- **Total Time Period ( $T$ ) and Frequency ( $f_0$ ):**

$$T = t_{ON} + t_{OFF} = 0.693(R_A + 2R_B)C \quad (1.8)$$

$$f_0 = \frac{1}{T} = \frac{1.44}{(R_A + 2R_B)C} \quad (1.9)$$

1. Determine resistor ratio for  $D = 0.60$ :

$$R_A = \left( \frac{2(0.60) - 1}{1 - 0.60} \right) R_B = \left( \frac{0.20}{0.40} \right) R_B = 0.5R_B \implies R_B = 2R_A \quad (1.10)$$

2. Express total period  $T$  in terms of  $R_B$ :

$$T = 0.693(0.5R_B + 2R_B)C = 0.693(2.5R_B)C = 1.7325 \cdot R_B C \quad (1.11)$$

3. Calculate target  $R_B$  and  $R_A$ :

$$R_B = \frac{T}{1.7325 \cdot C} = \frac{1.0 \times 10^{-3} \text{ s}}{1.7325 \times (0.1 \times 10^{-6} \text{ F})} \approx 5772 \text{ } \Omega = 5.77 \text{ k}\Omega \quad (1.12)$$

$$R_A = 0.5 \cdot R_B = 2.89 \text{ k}\Omega \quad (1.13)$$

4. Select nearest standard E12 resistor values:

$$R_A = 2.7 \text{ k}\Omega, \quad R_B = 5.6 \text{ k}\Omega \quad (1.14)$$

5. Recalculate the expected parameters with standard components:

$$t_1 = 0.693 \times (2.7 \text{ k}\Omega + 5.6 \text{ k}\Omega) \times 0.1 \text{ } \mu\text{F} = 0.575 \text{ ms} \quad (1.15)$$

$$t_2 = 0.693 \times (5.6 \text{ k}\Omega) \times 0.1 \text{ } \mu\text{F} = 0.388 \text{ ms} \quad (1.16)$$

$$T_{\text{design}} = 0.575 \text{ ms} + 0.388 \text{ ms} = 0.963 \text{ ms} \quad (1.17)$$

$$f_{0,\text{design}} = \frac{1}{0.963 \times 10^{-3} \text{ s}} \approx 1038.4 \text{ Hz} \quad (1.04 \text{ kHz}) \quad (1.18)$$

$$D_{\text{design}} = \frac{2.7 + 5.6}{2.7 + 2(5.6)} \times 100\% = \frac{8.3}{13.9} \times 100\% = 59.71\% \approx 60\% \quad (1.19)$$

## 1.7 Procedure

1. Assemble the circuit on a breadboard using  $R_A = 2.7 \text{ k}\Omega$ ,  $R_B = 5.6 \text{ k}\Omega$ , and  $C = 0.1 \text{ } \mu\text{F}$  according to Figure 1.1.
2. Connect a bypass capacitor  $C_2 = 0.01 \text{ } \mu\text{F}$  between Pin 5 and GND to suppress noise pickup.
3. Apply +5 V DC power to Pin 8 and Pin 4, and ground Pin 1.
4. Connect Channel 1 of the DSO to  $V_C$  (Pins 2/6) and Channel 2 to  $V_{out}$  (Pin 3).
5. Set DSO channels to DC coupling and adjust timebase to display 2 to 3 full cycles.
6. Measure high time  $t_1$ , low time  $t_2$ , upper threshold  $V_{TH}$ , and lower threshold  $V_{TL}$ .
7. Calculate practical period  $T = t_1 + t_2$ , frequency  $f_0 = 1/T$ , and duty cycle  $D = (t_1/T) \times 100\%$ .

## 1.8 Precautions

1. Ensure Pin 4 (RESET) is connected directly to  $+V_{CC}$  to avoid spontaneous resetting.
2. Use non-polarized ceramic or film capacitors for timing capacitor  $C$ .
3. Ensure all ground leads from power supply and DSO probes share a single common ground node.

## 1.9 Model Graph / Expected Waveforms

Figure 1.2 illustrates the expected time-aligned output square wave ( $V_{out}$ ) and capacitor exponential charge/discharge waveform ( $V_C$ ) for a 60% duty cycle design.

### Synchronized Waveforms for IC 555 Astable Multivibrator ( $D \approx 60\%$ )

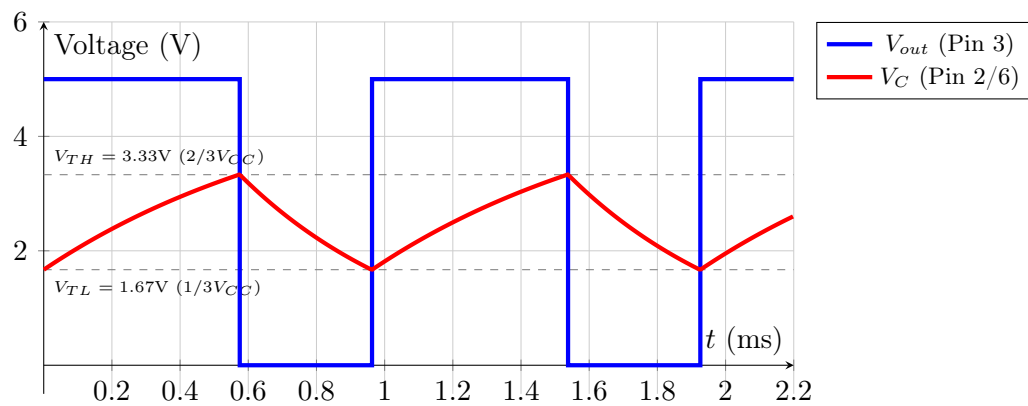


Figure 1.2: Model Output and Capacitor Voltage Waveforms for 60% Duty Ratio Design

Table 1.2: Theoretical vs Practical Waveform Measurements

| Parameter           | Theoretical Value | Measured Value | % Error |
|---------------------|-------------------|----------------|---------|
| High Time ( $t_1$ ) | 0.575 ms          |                |         |
| Low Time ( $t_2$ )  | 0.388 ms          |                |         |
| Time Period ( $T$ ) | 0.963 ms          |                |         |
| Frequency ( $f_0$ ) | 1038.4 Hz         |                |         |
| Duty Cycle ( $D$ )  | 59.71%            |                |         |

## 1.10 Observation Tables

### 1.10.1 Waveform Parameters

Record timing and frequency parameters in Table 1.2.

### 1.11 Result

The IC 555 astable multivibrator was designed and tested for a target 60% duty ratio:

- Measured high time  $t_1 =$  \_\_\_\_\_ ms and low time  $t_2 =$  \_\_\_\_\_ ms, giving a practical time period  $T =$  \_\_\_\_\_ ms.
- The experimental duty cycle was calculated as  $D_{\text{practical}} =$  \_\_\_\_\_% (Theoretical: 59.71%, target: 60%).
- Output oscillation frequency was measured as  $f_{0,\text{practical}} =$  \_\_\_\_\_ Hz compared to the theoretical value of 1038.4 Hz.

### 1.12 Questions

1. How can a steering diode connected across  $R_B$  enable independent adjustment of duty cycles below 50%?
2. What are the upper and lower limits on duty cycle  $D$  achievable with the standard 555 astable topology without steering diodes?
3. How can an astable 555 multivibrator circuit be modified to get a 50% duty cycle square wave?

4. How does replacing  $R_A$  and  $R_B$  with a potentiometer affect the frequency and duty cycle?
5. What is the effect on output duty cycle if supply voltage  $V_{CC}$  fluctuates between +5 V and +12 V?
6. What happens to the oscillation frequency and output amplitude if the supply voltage  $V_{CC}$  is increased from +5 V to +12 V? Explain why.
7. What is the functional role of Pin 5 (Control Voltage), and how can it be used to achieve frequency modulation (FM)?
8. Why does the timing capacitor  $C$  charge through  $(R_A + R_B)$  but discharge only through  $R_B$ ?
9. Compare the IC 555 astable multivibrator with an op-amp relaxation oscillator in terms of frequency stability and output driving capability.

## Appendix A

# Manufacturer Datasheets

**NPN general purpose transistors****BC107; BC108; BC109****FEATURES**

- Low current (max. 100 mA)
- Low voltage (max. 45 V).

**APPLICATIONS**

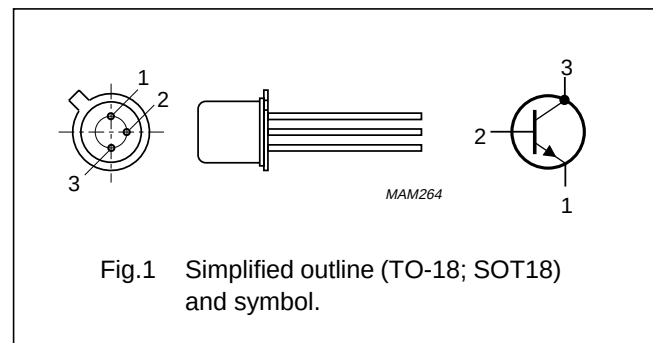
- General purpose switching and amplification.

**DESCRIPTION**

NPN transistor in a TO-18; SOT18 metal package.  
PNP complement: BC177.

**PINNING**

| PIN | DESCRIPTION                      |
|-----|----------------------------------|
| 1   | emitter                          |
| 2   | base                             |
| 3   | collector, connected to the case |

**QUICK REFERENCE DATA**

| SYMBOL    | PARAMETER                                          | CONDITIONS                                                    | MIN. | MAX. | UNIT |
|-----------|----------------------------------------------------|---------------------------------------------------------------|------|------|------|
| $V_{CBO}$ | collector-base voltage<br>BC107<br>BC108; BC109    | open emitter                                                  | —    | 50   | V    |
|           |                                                    |                                                               | —    | 30   | V    |
| $V_{CEO}$ | collector-emitter voltage<br>BC107<br>BC108; BC109 | open base                                                     | —    | 45   | V    |
|           |                                                    |                                                               | —    | 20   | V    |
| $I_{CM}$  | peak collector current                             |                                                               | —    | 200  | mA   |
| $P_{tot}$ | total power dissipation                            | $T_{amb} \leq 25\text{ °C}$                                   | —    | 300  | mW   |
| $h_{FE}$  | DC current gain<br>BC107<br>BC108<br>BC109         | $I_C = 2\text{ mA}; V_{CE} = 5\text{ V}$                      | 110  | 450  |      |
|           |                                                    |                                                               | 110  | 800  |      |
|           |                                                    |                                                               | 200  | 800  |      |
|           |                                                    |                                                               |      |      |      |
| $f_T$     | transition frequency                               | $I_C = 10\text{ mA}; V_{CE} = 5\text{ V}; f = 100\text{ MHz}$ | 100  | —    | MHz  |

## NPN general purpose transistors

## BC107; BC108; BC109

**LIMITING VALUES**

In accordance with the Absolute Maximum Rating System (IEC 134).

| SYMBOL    | PARAMETER                     | CONDITIONS                  | MIN. | MAX. | UNIT |
|-----------|-------------------------------|-----------------------------|------|------|------|
| $V_{CBO}$ | collector-base voltage        | open emitter                |      |      |      |
|           | BC107                         |                             | –    | 50   | V    |
|           | BC108; BC109                  |                             | –    | 30   | V    |
| $V_{CEO}$ | collector-emitter voltage     | open base                   |      |      |      |
|           | BC107                         |                             | –    | 45   | V    |
|           | BC108; BC109                  |                             | –    | 20   | V    |
| $V_{EBO}$ | emitter-base voltage          | open collector              |      |      |      |
|           | BC107                         |                             | –    | 6    | V    |
|           | BC108; BC109                  |                             | –    | 5    | V    |
| $I_C$     | collector current (DC)        |                             | –    | 100  | mA   |
| $I_{CM}$  | peak collector current        |                             | –    | 200  | mA   |
| $I_{BM}$  | peak base current             |                             | –    | 200  | mA   |
| $P_{tot}$ | total power dissipation       | $T_{amb} \leq 25\text{ °C}$ | –    | 300  | mW   |
| $T_{stg}$ | storage temperature           |                             | –65  | +150 | °C   |
| $T_j$     | junction temperature          |                             | –    | 175  | °C   |
| $T_{amb}$ | operating ambient temperature |                             | –65  | +150 | °C   |

**THERMAL CHARACTERISTICS**

| SYMBOL        | PARAMETER                                   | CONDITIONS | VALUE | UNIT |
|---------------|---------------------------------------------|------------|-------|------|
| $R_{th\ j-a}$ | thermal resistance from junction to ambient | note 1     | 0.5   | K/mW |
| $R_{th\ j-c}$ | thermal resistance from junction to case    |            | 0.2   | K/mW |

**Note**

1. Transistor mounted on an FR4 printed-circuit board.

## NPN general purpose transistors

## BC107; BC108; BC109

## CHARACTERISTICS

$T_j = 25\text{ }^{\circ}\text{C}$  unless otherwise specified.

| SYMBOL      | PARAMETER                                                                     | CONDITIONS                                                                                                            | MIN. | TYP. | MAX. | UNIT          |
|-------------|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|------|------|------|---------------|
| $I_{CBO}$   | collector cut-off current                                                     | $I_E = 0; V_{CB} = 20\text{ V}$                                                                                       | –    | –    | 15   | nA            |
|             |                                                                               | $I_E = 0; V_{CB} = 20\text{ V}; T_j = 150\text{ }^{\circ}\text{C}$                                                    | –    | –    | 15   | $\mu\text{A}$ |
| $I_{EBO}$   | emitter cut-off current                                                       | $I_C = 0; V_{EB} = 5\text{ V}$                                                                                        | –    | –    | 50   | nA            |
| $h_{FE}$    | DC current gain<br>BC107A; BC108A<br>BC107B; BC108B; BC109B<br>BC108C; BC109C | $I_C = 10\text{ }\mu\text{A}; V_{CE} = 5\text{ V}$                                                                    | –    | 90   | –    |               |
|             |                                                                               |                                                                                                                       | 40   | 150  | –    |               |
|             |                                                                               |                                                                                                                       | 100  | 270  | –    |               |
|             |                                                                               |                                                                                                                       |      |      |      |               |
| $h_{FE}$    | DC current gain<br>BC107A; BC108A<br>BC107B; BC108B; BC109B<br>BC108C; BC109C | $I_C = 2\text{ mA}; V_{CE} = 5\text{ V}$                                                                              | 110  | 180  | 220  |               |
|             |                                                                               |                                                                                                                       | 200  | 290  | 450  |               |
|             |                                                                               |                                                                                                                       | 420  | 520  | 800  |               |
|             |                                                                               |                                                                                                                       |      |      |      |               |
| $V_{CEsat}$ | collector-emitter saturation voltage                                          | $I_C = 10\text{ mA}; I_B = 0.5\text{ mA}$                                                                             | –    | 90   | 250  | mV            |
|             |                                                                               | $I_C = 100\text{ mA}; I_B = 5\text{ mA}$                                                                              | –    | 200  | 600  | mV            |
| $V_{BEsat}$ | base-emitter saturation voltage                                               | $I_C = 10\text{ mA}; I_B = 0.5\text{ mA}; \text{note 1}$                                                              | –    | 700  | –    | mV            |
|             |                                                                               | $I_C = 100\text{ mA}; I_B = 5\text{ mA}; \text{note 1}$                                                               | –    | 900  | –    | mV            |
| $V_{BE}$    | base-emitter voltage                                                          | $I_C = 2\text{ mA}; V_{CE} = 5\text{ V}; \text{note 2}$                                                               | 550  | 620  | 700  | mV            |
|             |                                                                               | $I_C = 10\text{ mA}; V_{CE} = 5\text{ V}; \text{note 2}$                                                              | –    | –    | 770  | mV            |
| $C_c$       | collector capacitance                                                         | $I_E = i_e = 0; V_{CB} = 10\text{ V}; f = 1\text{ MHz}$                                                               | –    | 2.5  | 6    | pF            |
| $C_e$       | emitter capacitance                                                           | $I_C = i_c = 0; V_{EB} = 0.5\text{ V}; f = 1\text{ MHz}$                                                              | –    | 9    | –    | pF            |
| $f_T$       | transition frequency                                                          | $I_C = 10\text{ mA}; V_{CB} = 5\text{ V}; f = 100\text{ MHz}$                                                         | 100  | –    | –    | MHz           |
| F           | noise figure<br>BC109B; BC109C                                                | $I_C = 200\text{ }\mu\text{A}; V_{CE} = 5\text{ V}; R_S = 2\text{ k}\Omega;$<br>$f = 30\text{ Hz to }15.7\text{ kHz}$ | –    | –    | 4    | dB            |
|             |                                                                               |                                                                                                                       |      |      |      |               |
| F           | noise figure<br>BC107A; BC108A<br>BC107B; BC108B; BC108C<br>BC109B; BC109C    | $I_C = 200\text{ }\mu\text{A}; V_{CE} = 5\text{ V}; R_S = 2\text{ k}\Omega;$<br>$f = 1\text{ kHz}; B = 200\text{ Hz}$ | –    | –    | 10   | dB            |
|             |                                                                               |                                                                                                                       | –    | –    | 4    | dB            |

## Notes

- $V_{BEsat}$  decreases by about 1.7 mV/K with increasing temperature.
- $V_{BE}$  decreases by about 2 mV/K with increasing temperature.



# 2N7000

## N-Channel Enhancement-Mode Vertical DMOS FET

### Features

- Free from Secondary Breakdown
- Low Power Drive Requirement
- Ease of Paralleling
- Low  $C_{ISS}$  and Fast Switching Speeds
- Excellent Thermal Stability
- Integral Source-Drain Diode
- High Input Impedance and High Gain

### Applications

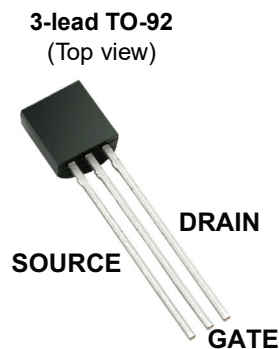
- Motor Controls
- Converters
- Amplifiers
- Switches
- Power Supply Circuits
- Drivers (Relays, Hammers, Solenoids, Lamps, Memory, Displays, Bipolar Transistors, etc.)

### General Description

The 2N7000 is an Enhancement-mode (normally-off) transistor that uses a vertical DMOS structure and a well-proven silicon gate manufacturing process. This combination produces a device with the power handling capabilities of bipolar transistors as well as the high input impedance and positive temperature coefficient inherent in MOS devices. Characteristic of all MOS structures, this device is free from thermal runaway and thermally induced secondary breakdown.

Microchip's vertical DMOS FETs are ideally suited to a wide range of switching and amplifying applications where very low threshold voltage, high breakdown voltage, high input impedance, low input capacitance, and fast switching speeds are desired.

### Package Type



See [Table 3-1](#) for pin information.

## 1.0 ELECTRICAL CHARACTERISTICS

### Absolute Maximum Ratings†

|                                            |                                   |
|--------------------------------------------|-----------------------------------|
| Drain-to-Source Voltage .....              | $BV_{DSS}$                        |
| Drain-to-Gate Voltage .....                | $BV_{DGS}$                        |
| Gate-to-Source Voltage .....               | $\pm 30V$                         |
| Operating Ambient Temperature, $T_A$ ..... | $-55^{\circ}C$ to $+150^{\circ}C$ |
| Storage Temperature, $T_S$ .....           | $-55^{\circ}C$ to $+150^{\circ}C$ |

† **Notice:** Stresses above those listed under “Absolute Maximum Ratings” may cause permanent damage to the device. This is a stress rating only, and functional operation of the device at those or any other conditions above those indicated in the operational sections of this specification is not intended. Exposure to maximum rating conditions for extended periods may affect device reliability.

### DC ELECTRICAL CHARACTERISTICS

**Electrical Specifications:**  $T_A = 25^{\circ}C$  unless otherwise specified. All DC parameters are 100% tested at  $25^{\circ}C$  unless otherwise stated. (Pulse test: 300  $\mu s$  pulse, 2% duty cycle)

| Parameter                                  | Sym.         | Min. | Typ. | Max. | Unit     | Conditions                                                  |
|--------------------------------------------|--------------|------|------|------|----------|-------------------------------------------------------------|
| Drain-to-Source Breakdown Voltage          | $BV_{DSS}$   | 60   | —    | —    | V        | $V_{GS} = 0V, I_D = 10 \mu A$                               |
| Gate Threshold Voltage                     | $V_{GS(th)}$ | 0.8  | —    | 3    | V        | $V_{GS} = V_{DS}, I_D = 1 mA$                               |
| Gate Body Leakage Current                  | $I_{GSS}$    | —    | —    | 10   | nA       | $V_{GS} = \pm 15V, V_{DS} = 0V$                             |
| Zero-Gate Voltage Drain Current            | $I_{DSS}$    | —    | —    | 1    | $\mu A$  | $V_{GS} = 0V, V_{DS} = 48V$                                 |
|                                            |              | —    | —    | 1    | mA       | $V_{GS} = 0V, V_{DS} = 48V, T_A = 125^{\circ}C$<br>(Note 1) |
| On-State Drain Current                     | $I_{D(ON)}$  | 75   | —    | —    | mA       | $V_{GS} = 4.5V, V_{DS} = 10V$                               |
| Static Drain-to-Source On-State Resistance | $R_{DS(ON)}$ | —    | —    | 5.3  | $\Omega$ | $V_{GS} = 4.5V, I_D = 75 mA$                                |
|                                            |              | —    | —    | 5    | $\Omega$ | $V_{GS} = 10V, I_D = 500 mA$                                |

**Note 1:** Specification is obtained by characterization and is not 100% tested.

### AC ELECTRICAL CHARACTERISTICS

**Electrical Specifications:**  $T_A = 25^{\circ}C$  unless otherwise specified. All AC parameters are not 100% sample tested.

| Parameter                        | Sym.         | Min. | Typ. | Max. | Unit | Conditions                                            |
|----------------------------------|--------------|------|------|------|------|-------------------------------------------------------|
| Forward Transconductance         | $G_{FS}$     | 100  | —    | —    | mmho | $V_{DS} = 10V, I_D = 200 mA$                          |
| Input Capacitance                | $C_{ISS}$    | —    | —    | 60   | pF   | $V_{GS} = 0V, V_{DS} = 25V,$<br>$f = 1 MHz$           |
| Common-Source Output Capacitance | $C_{OSS}$    | —    | —    | 25   | pF   |                                                       |
| Reverse Transfer Capacitance     | $C_{RSS}$    | —    | —    | 5    | pF   |                                                       |
| Turn-On Delay Time               | $t_{d(ON)}$  | —    | —    | 10   | ns   | $V_{DD} = 15V, I_D = 500 mA,$<br>$R_{GEN} = 25\Omega$ |
| Turn-Off Delay Time              | $t_{d(OFF)}$ | —    | —    | 10   | ns   |                                                       |

#### DIODE PARAMETER

|                            |          |   |      |   |   |                                         |
|----------------------------|----------|---|------|---|---|-----------------------------------------|
| Diode Forward Voltage Drop | $V_{SD}$ | — | 0.85 | — | V | $V_{GS} = 0V, I_{SD} = 200 mA$ (Note 1) |
|----------------------------|----------|---|------|---|---|-----------------------------------------|

**Note 1:** All DC parameters are 100% tested at  $25^{\circ}C$  unless otherwise stated.  
(Pulse test: 300  $\mu s$  pulse, 2% duty cycle)

**TEMPERATURE SPECIFICATIONS**

| <b>Electrical Characteristics:</b> Unless otherwise specified, for all specifications $T_A = T_J = +25^\circ\text{C}$ . |               |      |      |      |                    |            |
|-------------------------------------------------------------------------------------------------------------------------|---------------|------|------|------|--------------------|------------|
| Parameter                                                                                                               | Sym.          | Min. | Typ. | Max. | Unit               | Conditions |
| <b>TEMPERATURE RANGE</b>                                                                                                |               |      |      |      |                    |            |
| Operating Ambient Temperature                                                                                           | $T_A$         | -55  | —    | +150 | $^\circ\text{C}$   |            |
| Storage Temperature                                                                                                     | $T_S$         | -55  | —    | +150 | $^\circ\text{C}$   |            |
| <b>PACKAGE THERMAL RESISTANCE</b>                                                                                       |               |      |      |      |                    |            |
| 3-lead TO-92                                                                                                            | $\theta_{JA}$ | —    | 132  | —    | $^\circ\text{C/W}$ |            |

**Note 1:** Mounted on an FR4 board; 25 mm x 25 mm x 1.57 mm

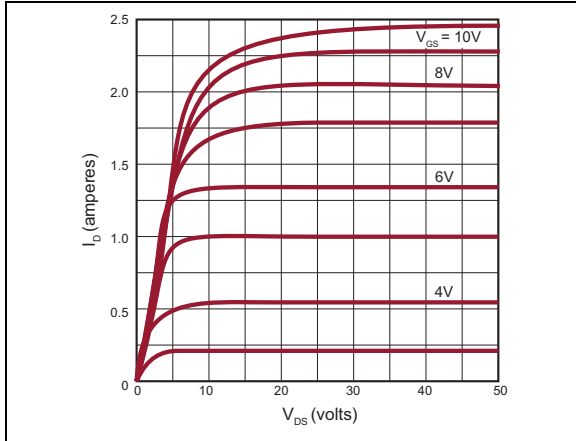
**THERMAL CHARACTERISTICS**

| Package      | $I_D$ (Note 1)<br>(Continuous)<br>(mA) | $I_D$<br>(Pulsed)<br>(mA) | Power Dissipation at<br>$T_C = 25^\circ\text{C}$<br>(W) | $I_{DR}$ (Note 1)<br>(mA) | $I_{DRM}$<br>(mA) |
|--------------|----------------------------------------|---------------------------|---------------------------------------------------------|---------------------------|-------------------|
| 3-lead TO-92 | 200                                    | 500                       | 1                                                       | 200                       | 500               |

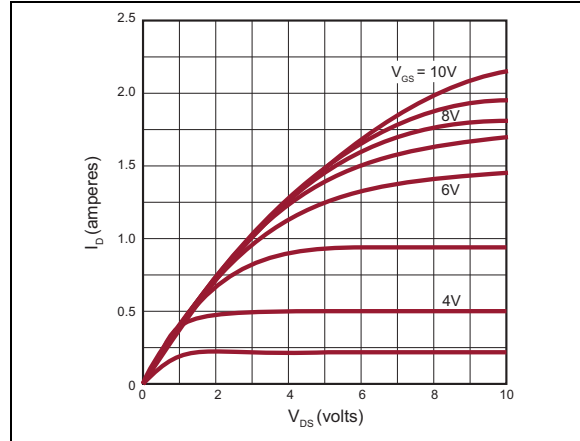
**Note 1:**  $I_D$  (continuous) is limited by maximum  $T_J$ .

## 2.0 TYPICAL PERFORMANCE CURVES

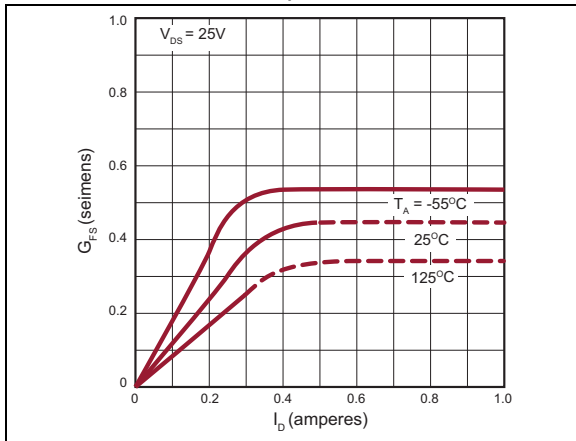
**Note:** The graphs and tables provided following this note are a statistical summary based on a limited number of samples and are provided for informational purposes only. The performance characteristics listed herein are not tested or guaranteed. In some graphs or tables, the data presented may be outside the specified operating range (e.g. outside specified power supply range) and therefore outside the warranted range.



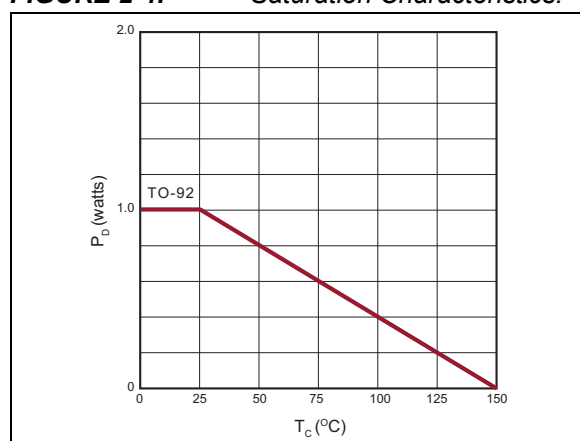
**FIGURE 2-1:** Output Characteristics.



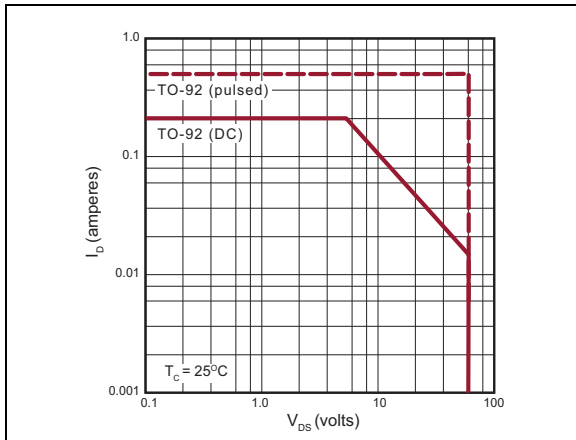
**FIGURE 2-4:** Saturation Characteristics.



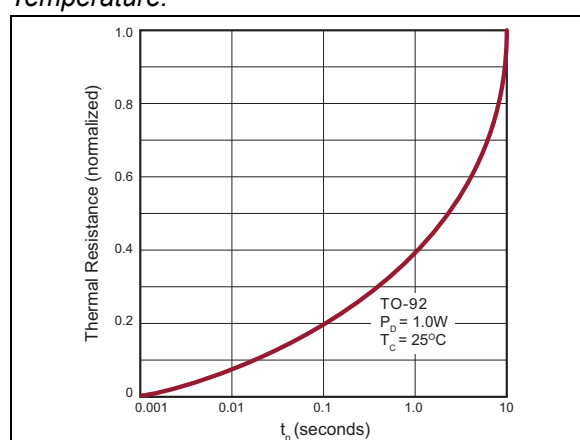
**FIGURE 2-2:** Transconductance vs. Drain Current.



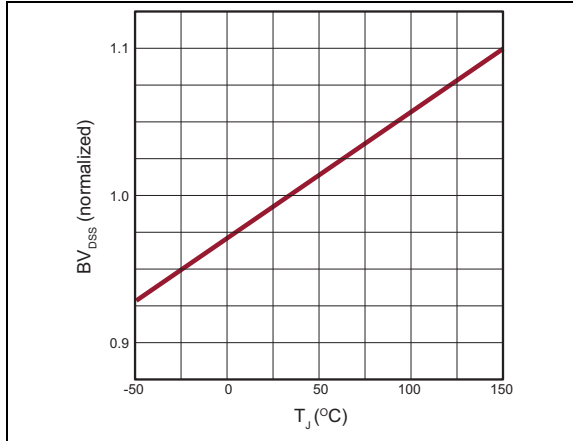
**FIGURE 2-5:** Power Dissipation vs. Case Temperature.



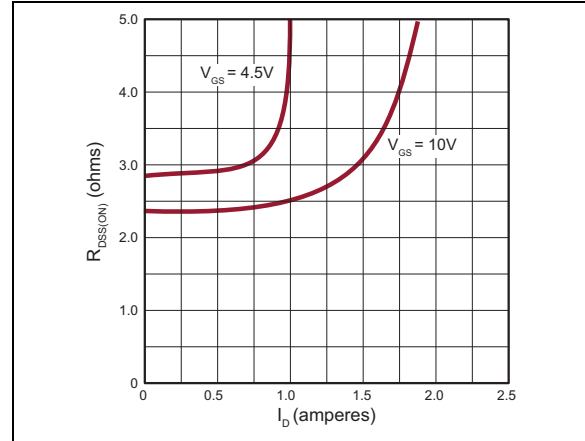
**FIGURE 2-3:** Maximum Rated Safe Operating Area.



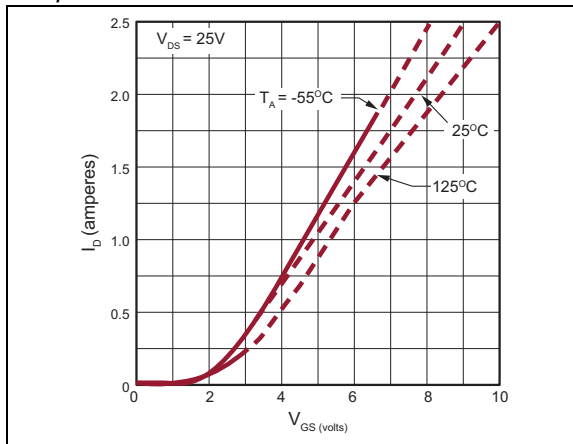
**FIGURE 2-6:** Thermal Response Characteristics.



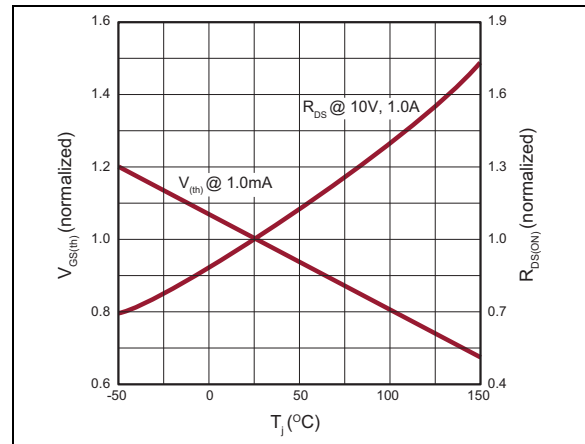
**FIGURE 2-7:**  $BV_{DSS}$  Variation with Temperature.



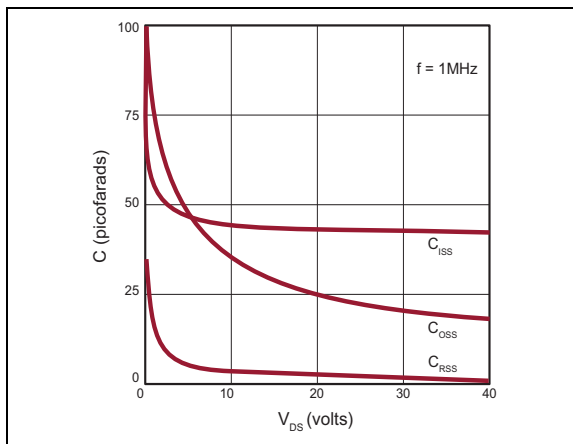
**FIGURE 2-10:** On-Resistance vs. Drain Current.



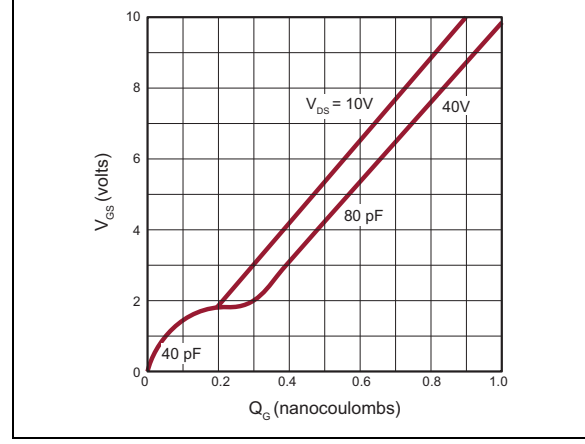
**FIGURE 2-8:** Transfer Characteristics.



**FIGURE 2-11:**  $V_{GS(th)}$  and  $R_{DS}$  Variation with Temperature.

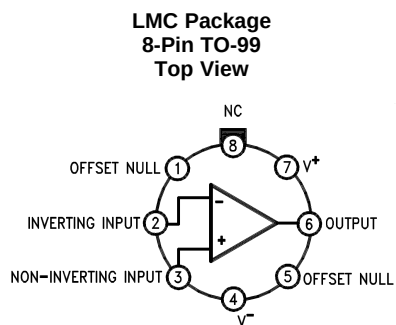


**FIGURE 2-9:** Capacitance vs. Drain-to-Source Voltage.

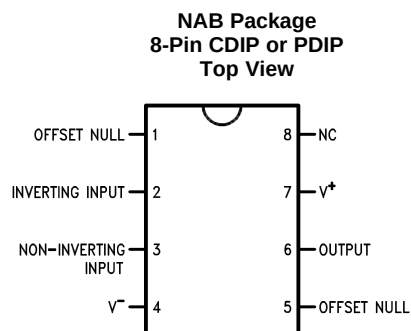


**FIGURE 2-12:** Gate Drive Dynamic Characteristics.

## 5 Pin Configuration and Functions



LM741H is available per JM38510/10101



### Pin Functions

| PIN                |      | I/O | DESCRIPTION                                                                          |
|--------------------|------|-----|--------------------------------------------------------------------------------------|
| NAME               | NO.  |     |                                                                                      |
| INVERTING INPUT    | 2    | I   | Inverting signal input                                                               |
| NC                 | 8    | N/A | No Connect, should be left floating                                                  |
| NONINVERTING INPUT | 3    | I   | Noninverting signal input                                                            |
| OFFSET NULL        | 1, 5 | I   | Offset null pin used to eliminate the offset voltage and balance the input voltages. |
| OFFSET NULL        |      |     |                                                                                      |
| OUTPUT             | 6    | O   | Amplified signal output                                                              |
| V+                 | 7    | I   | Positive supply voltage                                                              |
| V-                 | 4    | I   | Negative supply voltage                                                              |

## 6 Specifications

### 6.1 Absolute Maximum Ratings

over operating free-air temperature range (unless otherwise noted)<sup>(1)(2)(3)</sup>

|                                       |                                    | MIN | MAX        | UNIT |
|---------------------------------------|------------------------------------|-----|------------|------|
| Supply voltage                        | LM741, LM741A                      |     | ±22        | V    |
|                                       | LM741C                             |     | ±18        |      |
| Power dissipation <sup>(4)</sup>      |                                    |     | 500        | mW   |
| Differential input voltage            |                                    |     | ±30        | V    |
| Input voltage <sup>(5)</sup>          |                                    |     | ±15        | V    |
| Output short circuit duration         |                                    |     | Continuous |      |
| Operating temperature                 | LM741, LM741A                      | –50 | 125        | °C   |
|                                       | LM741C                             | 0   | 70         |      |
| Junction temperature                  | LM741, LM741A                      |     | 150        | °C   |
|                                       | LM741C                             |     | 100        |      |
| Soldering information                 | PDIP package (10 seconds)          |     | 260        | °C   |
|                                       | CDIP or TO-99 package (10 seconds) |     | 300        | °C   |
| Storage temperature, T <sub>stg</sub> |                                    | –65 | 150        | °C   |

- (1) Stresses beyond those listed under *Absolute Maximum Ratings* may cause permanent damage to the device. These are stress ratings only, which do not imply functional operation of the device at these or any other conditions beyond those indicated under *Recommended Operating Conditions*. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.
- (2) For military specifications see RETS741X for LM741 and RETS741AX for LM741A.
- (3) If Military/Aerospace specified devices are required, please contact the TI Sales Office/Distributors for availability and specifications.
- (4) For operation at elevated temperatures, these devices must be derated based on thermal resistance, and T<sub>j</sub> max. (listed under “Absolute Maximum Ratings”).  $T_j = T_A + (\theta_{JA} P_D)$ .
- (5) For supply voltages less than ±15 V, the absolute maximum input voltage is equal to the supply voltage.

### 6.2 ESD Ratings

|                    |                                                                                           | VALUE | UNIT |
|--------------------|-------------------------------------------------------------------------------------------|-------|------|
| V <sub>(ESD)</sub> | Electrostatic discharge Human body model (HBM), per ANSI/ESDA/JEDEC JS-001 <sup>(1)</sup> | ±400  | V    |

- (1) Level listed above is the passing level per ANSI, ESDA, and JEDEC JS-001. JEDEC document JEP155 states that 500-V HBM allows safe manufacturing with a standard ESD control process.

### 6.3 Recommended Operating Conditions

over operating free-air temperature range (unless otherwise noted)

|                          |               | MIN | NOM | MAX | UNIT |
|--------------------------|---------------|-----|-----|-----|------|
| Supply voltage (VDD-GND) | LM741, LM741A | ±10 | ±15 | ±22 | V    |
|                          | LM741C        | ±10 | ±15 | ±18 |      |
| Temperature              | LM741, LM741A | –55 |     | 125 | °C   |
|                          | LM741C        | 0   |     | 70  |      |

### 6.4 Thermal Information

| THERMAL METRIC <sup>(1)</sup> |                                           | LM741       |            |          | UNIT |
|-------------------------------|-------------------------------------------|-------------|------------|----------|------|
|                               |                                           | LMC (TO-99) | NAB (CDIP) | P (PDIP) |      |
|                               |                                           | 8 PINS      | 8 PINS     | 8 PINS   |      |
| R <sub>θJA</sub>              | Junction-to-ambient thermal resistance    | 170         | 100        | 100      | °C/W |
| R <sub>θJC(top)</sub>         | Junction-to-case (top) thermal resistance | 25          | —          | —        | °C/W |

- (1) For more information about traditional and new thermal metrics, see the *Semiconductor and IC Package Thermal Metrics* application report, [SPRA953](#).

## 6.5 Electrical Characteristics, LM741<sup>(1)</sup>

| PARAMETER                             |           | TEST CONDITIONS                                                                                                             |                                                 | MIN | TYP      | MAX      | UNIT             |
|---------------------------------------|-----------|-----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|-----|----------|----------|------------------|
| Input offset voltage                  |           | $R_S \leq 10\text{ k}\Omega$                                                                                                | $T_A = 25^\circ\text{C}$                        |     | 1        | 5        | mV               |
|                                       |           |                                                                                                                             | $T_{\text{AMIN}} \leq T_A \leq T_{\text{AMAX}}$ |     |          | 6        | mV               |
| Input offset voltage adjustment range |           | $T_A = 25^\circ\text{C}, V_S = \pm 20\text{ V}$                                                                             |                                                 |     | $\pm 15$ |          | mV               |
| Input offset current                  |           | $T_A = 25^\circ\text{C}$                                                                                                    |                                                 |     | 20       | 200      | nA               |
|                                       |           | $T_{\text{AMIN}} \leq T_A \leq T_{\text{AMAX}}$                                                                             |                                                 |     | 85       | 500      |                  |
| Input bias current                    |           | $T_A = 25^\circ\text{C}$                                                                                                    |                                                 |     | 80       | 500      | nA               |
|                                       |           | $T_{\text{AMIN}} \leq T_A \leq T_{\text{AMAX}}$                                                                             |                                                 |     |          | 1.5      | $\mu\text{A}$    |
| Input resistance                      |           | $T_A = 25^\circ\text{C}, V_S = \pm 20\text{ V}$                                                                             |                                                 |     | 0.3      | 2        | M $\Omega$       |
| Input voltage range                   |           | $T_{\text{AMIN}} \leq T_A \leq T_{\text{AMAX}}$                                                                             |                                                 |     | $\pm 12$ | $\pm 13$ | V                |
| Large signal voltage gain             |           | $V_S = \pm 15\text{ V}, V_O = \pm 10\text{ V}, R_L \geq 2\text{ k}\Omega$                                                   | $T_A = 25^\circ\text{C}$                        |     | 50       | 200      | V/mV             |
|                                       |           |                                                                                                                             | $T_{\text{AMIN}} \leq T_A \leq T_{\text{AMAX}}$ |     | 25       |          |                  |
| Output voltage swing                  |           | $V_S = \pm 15\text{ V}$                                                                                                     | $R_L \geq 10\text{ k}\Omega$                    |     | $\pm 12$ | $\pm 14$ | V                |
|                                       |           |                                                                                                                             | $R_L \geq 2\text{ k}\Omega$                     |     | $\pm 10$ | $\pm 13$ |                  |
| Output short circuit current          |           | $T_A = 25^\circ\text{C}$                                                                                                    |                                                 |     | 25       |          | mA               |
| Common-mode rejection ratio           |           | $R_S \leq 10\text{ }\Omega, V_{\text{CM}} = \pm 12\text{ V}, T_{\text{AMIN}} \leq T_A \leq T_{\text{AMAX}}$                 |                                                 |     | 80       | 95       | dB               |
| Supply voltage rejection ratio        |           | $V_S = \pm 20\text{ V}$ to $V_S = \pm 5\text{ V}, R_S \leq 10\text{ }\Omega, T_{\text{AMIN}} \leq T_A \leq T_{\text{AMAX}}$ |                                                 |     | 86       | 96       | dB               |
| Transient response                    | Rise time | $T_A = 25^\circ\text{C}, \text{unity gain}$                                                                                 |                                                 |     | 0.3      |          | $\mu\text{s}$    |
|                                       | Overshoot |                                                                                                                             |                                                 |     | 5%       |          |                  |
| Slew rate                             |           | $T_A = 25^\circ\text{C}, \text{unity gain}$                                                                                 |                                                 |     | 0.5      |          | V/ $\mu\text{s}$ |
| Supply current                        |           | $T_A = 25^\circ\text{C}$                                                                                                    |                                                 |     | 1.7      | 2.8      | mA               |
| Power consumption                     |           | $V_S = \pm 15\text{ V}$                                                                                                     | $T_A = 25^\circ\text{C}$                        |     | 50       | 85       | mW               |
|                                       |           |                                                                                                                             | $T_A = T_{\text{AMIN}}$                         |     | 60       | 100      |                  |
|                                       |           |                                                                                                                             | $T_A = T_{\text{AMAX}}$                         |     | 45       | 75       |                  |

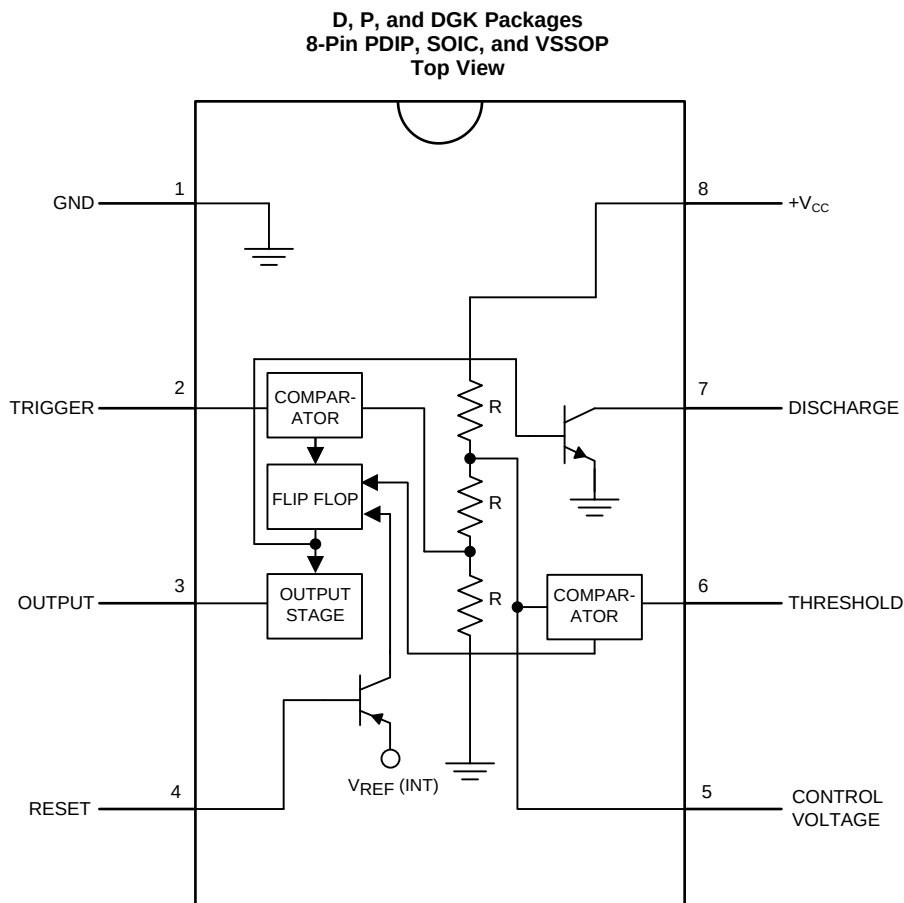
(1) Unless otherwise specified, these specifications apply for  $V_S = \pm 15 \text{ V}, -55^\circ\text{C} \leq T_A \leq +125^\circ\text{C}$  (LM741/LM741A). For the LM741C/LM741E, these specifications are limited to  $0^\circ\text{C} \leq T_A \leq +70^\circ\text{C}$ .

## 6.6 Electrical Characteristics, LM741A<sup>(1)</sup>

| PARAMETER                             | TEST CONDITIONS                                                                                             |                                   | MIN      | TYP | MAX  | UNIT                         |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------|-----------------------------------|----------|-----|------|------------------------------|
| Input offset voltage                  | $R_S \leq 50 \Omega$                                                                                        | $T_A = 25^\circ\text{C}$          |          | 0.8 | 3    | mV                           |
|                                       |                                                                                                             | $T_{AMIN} \leq T_A \leq T_{AMAX}$ |          |     | 4    | mV                           |
| Average input offset voltage drift    |                                                                                                             |                                   |          |     | 15   | $\mu\text{V}/^\circ\text{C}$ |
| Input offset voltage adjustment range | $T_A = 25^\circ\text{C}, V_S = \pm 20 \text{ V}$                                                            |                                   | $\pm 10$ |     |      | mV                           |
| Input offset current                  | $T_A = 25^\circ\text{C}$                                                                                    |                                   |          | 3   | 30   | nA                           |
|                                       | $T_{AMIN} \leq T_A \leq T_{AMAX}$                                                                           |                                   |          |     | 70   |                              |
| Average input offset current drift    |                                                                                                             |                                   |          |     | 0.5  | nA/ $^\circ\text{C}$         |
| Input bias current                    | $T_A = 25^\circ\text{C}$                                                                                    |                                   |          | 30  | 80   | nA                           |
|                                       | $T_{AMIN} \leq T_A \leq T_{AMAX}$                                                                           |                                   |          |     | 0.21 | $\mu\text{A}$                |
| Input resistance                      | $T_A = 25^\circ\text{C}, V_S = \pm 20 \text{ V}$                                                            |                                   | 1        | 6   |      | M $\Omega$                   |
|                                       | $T_{AMIN} \leq T_A \leq T_{AMAX}, V_S = \pm 20 \text{ V}$                                                   |                                   | 0.5      |     |      |                              |
| Large signal voltage gain             | $V_S = \pm 20 \text{ V}, V_O = \pm 15 \text{ V}, R_L \geq 2 \text{ k}\Omega$                                | $T_A = 25^\circ\text{C}$          | 50       |     |      | V/mV                         |
|                                       |                                                                                                             | $T_{AMIN} \leq T_A \leq T_{AMAX}$ | 32       |     |      |                              |
|                                       | $V_S = \pm 5 \text{ V}, V_O = \pm 2 \text{ V}, R_L \geq 2 \text{ k}\Omega, T_{AMIN} \leq T_A \leq T_{AMAX}$ |                                   | 10       |     |      |                              |

(1) Unless otherwise specified, these specifications apply for  $V_S = \pm 15 \text{ V}, -55^\circ\text{C} \leq T_A \leq +125^\circ\text{C}$  (LM741/LM741A). For the LM741C/LM741E, these specifications are limited to  $0^\circ\text{C} \leq T_A \leq +70^\circ\text{C}$ .

## 5 Pin Configuration and Functions



### Pin Functions

| PIN |                 | I/O | DESCRIPTION                                                                                                                                                                               |
|-----|-----------------|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NO. | NAME            |     |                                                                                                                                                                                           |
| 5   | Control Voltage | I   | Controls the threshold and trigger levels. It determines the pulse width of the output waveform. An external voltage applied to this pin can also be used to modulate the output waveform |
| 7   | Discharge       | I   | Open collector output which discharges a capacitor between intervals (in phase with output). It toggles the output from high to low when voltage reaches 2/3 of the supply voltage        |
| 1   | GND             | O   | Ground reference voltage                                                                                                                                                                  |
| 3   | Output          | O   | Output driven waveform                                                                                                                                                                    |
| 4   | Reset           | I   | Negative pulse applied to this pin to disable or reset the timer. When not used for reset purposes, it should be connected to VCC to avoid false triggering                               |
| 6   | Threshold       | I   | Compares the voltage applied to the terminal with a reference voltage of 2/3 Vcc. The amplitude of voltage applied to this terminal is responsible for the set state of the flip-flop     |
| 2   | Trigger         | I   | Responsible for transition of the flip-flop from set to reset. The output of the timer depends on the amplitude of the external trigger pulse applied to this pin                         |
| 8   | V <sup>+</sup>  | I   | Supply voltage with respect to GND                                                                                                                                                        |

## 6.5 Electrical Characteristics

(T<sub>A</sub> = 25°C, V<sub>CC</sub> = 5 V to 15 V, unless otherwise specified)<sup>(1)(2)</sup>

| PARAMETER                 | TEST CONDITIONS                                                          | MIN | TYP    | MAX  | UNIT              |
|---------------------------|--------------------------------------------------------------------------|-----|--------|------|-------------------|
| Supply Voltage            |                                                                          | 4.5 |        | 16   | V                 |
| Supply Current            | V <sub>CC</sub> = 5 V, R <sub>L</sub> = ∞                                |     | 3      | 6    | mA                |
|                           | V <sub>CC</sub> = 15 V, R <sub>L</sub> = ∞<br>(Low State) <sup>(3)</sup> |     | 10     | 15   |                   |
| Timing Error, Monostable  |                                                                          |     |        |      |                   |
| Initial Accuracy          |                                                                          |     | 1 %    |      |                   |
| Drift with Temperature    | R <sub>A</sub> = 1 k to 100 kΩ,                                          |     | 50     |      | ppm/°C            |
|                           | C = 0.1 μF, <sup>(4)</sup>                                               |     |        |      |                   |
| Accuracy over Temperature |                                                                          |     | 1.5 %  |      |                   |
| Drift with Supply         |                                                                          |     | 0.1 %  |      | V                 |
| Timing Error, Astable     |                                                                          |     |        |      |                   |
| Initial Accuracy          |                                                                          |     | 2.25   |      |                   |
| Drift with Temperature    | R <sub>A</sub> , R <sub>B</sub> = 1 k to 100 kΩ,                         |     | 150    |      | ppm/°C            |
|                           | C = 0.1 μF, <sup>(4)</sup>                                               |     |        |      |                   |
| Accuracy over Temperature |                                                                          |     | 3.0%   |      |                   |
| Drift with Supply         |                                                                          |     | 0.30 % |      | /V                |
| Threshold Voltage         |                                                                          |     | 0.667  |      | x V <sub>CC</sub> |
| Trigger Voltage           | V <sub>CC</sub> = 15 V                                                   |     | 5      |      | V                 |
|                           | V <sub>CC</sub> = 5 V                                                    |     | 1.67   |      | V                 |
| Trigger Current           |                                                                          |     | 0.5    | 0.9  | μA                |
| Reset Voltage             |                                                                          | 0.4 | 0.5    | 1    | V                 |
| Reset Current             |                                                                          |     | 0.1    | 0.4  | mA                |
| Threshold Current         | <sup>(5)</sup>                                                           |     | 0.1    | 0.25 | μA                |
| Control Voltage Level     | V <sub>CC</sub> = 15 V                                                   | 9   | 10     | 11   | V                 |
|                           | V <sub>CC</sub> = 5 V                                                    | 2.6 | 3.33   | 4    |                   |
| Pin 7 Leakage Output High |                                                                          |     | 1      | 100  | nA                |
| Pin 7 Sat <sup>(6)</sup>  |                                                                          |     |        |      |                   |
| Output Low                | V <sub>CC</sub> = 15 V, I <sub>L</sub> = 15 mA                           |     | 180    |      | mV                |
| Output Low                | V <sub>CC</sub> = 4.5 V, I <sub>L</sub> = 4.5 mA                         |     | 80     | 200  | mV                |
| Output Voltage Drop (Low) | V <sub>CC</sub> = 15 V                                                   |     |        |      |                   |
|                           | I <sub>SINK</sub> = 10 mA                                                |     | 0.1    | 0.25 | V                 |
|                           | I <sub>SINK</sub> = 50 mA                                                |     | 0.4    | 0.75 | V                 |
|                           | I <sub>SINK</sub> = 100 mA                                               |     | 2      | 2.5  | V                 |
|                           | I <sub>SINK</sub> = 200 mA                                               |     | 2.5    |      | V                 |
|                           | V <sub>CC</sub> = 5 V                                                    |     |        |      |                   |
|                           | I <sub>SINK</sub> = 8 mA                                                 |     |        |      | V                 |
|                           | I <sub>SINK</sub> = 5 mA                                                 |     | 0.25   | 0.35 | V                 |

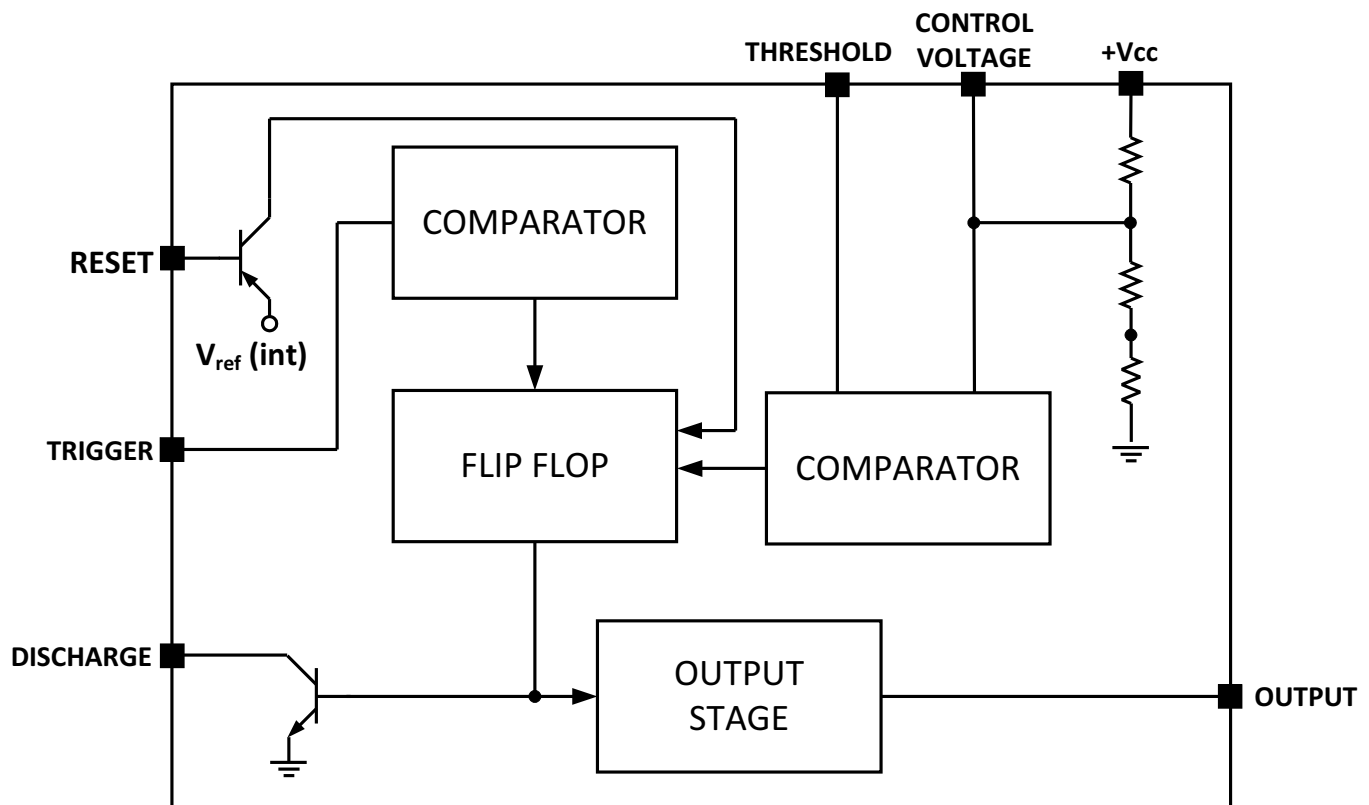
- (1) All voltages are measured with respect to the ground pin, unless otherwise specified.
- (2) [Absolute Maximum Ratings](#) indicate limits beyond which damage to the device may occur. [Recommended Operating Conditions](#) indicate conditions for which the device is functional, but do not ensure specific performance limits. [Electrical Characteristics](#) state DC and AC electrical specifications under particular test conditions which ensures specific performance limits. This assumes that the device is within the [Recommended Operating Conditions](#). Specifications are not ensured for parameters where no limit is given, however, the typical value is a good indication of device performance.
- (3) Supply current when output high typically 1 mA less at V<sub>CC</sub> = 5 V.
- (4) Tested at V<sub>CC</sub> = 5 V and V<sub>CC</sub> = 15 V.
- (5) This will determine the maximum value of R<sub>A</sub> + R<sub>B</sub> for 15 V operation. The maximum total (R<sub>A</sub> + R<sub>B</sub>) is 20 MΩ.
- (6) No protection against excessive pin 7 current is necessary providing the package dissipation rating will not be exceeded.

## 7 Detailed Description

### 7.1 Overview

The LM555 is a highly stable device for generating accurate time delays or oscillation. Additional terminals are provided for triggering or resetting if desired. In the time delay mode of operation, the time is precisely controlled by one external resistor and capacitor. For astable operation as an oscillator, the free running frequency and duty cycle are accurately controlled with two external resistors and one capacitor. The circuit may be triggered and reset on falling waveforms, and the output circuit can source or sink up to 200mA or driver TTL circuits. The LM555 are available in 8-pin PDIP, SOIC, and VSSOP packages and is a direct replacement for SE555/NE555.

### 7.2 Functional Block Diagram



### 7.3 Feature Description

#### 7.3.1 Direct Replacement for SE555/NE555

The LM555 timer is a direct replacement for SE555 and NE555. It is pin-to-pin compatible so that no schematic or layout changes are necessary. The LM555 come in an 8-pin PDIP, SOIC, and VSSOP package.

#### 7.3.2 Timing From Microseconds Through Hours

The LM555 has the ability to have timing parameters from the microseconds range to hours. The time delay of the system can be determined by the time constant of the R and C value used for either the monostable or astable configuration. A nomograph is available for easy determination of R and C values for various time delays.

#### 7.3.3 Operates in Both Astable and Monostable Mode

The LM555 can operate in both astable and monostable mode depending on the application requirements.

- Monostable mode: The LM555 timer acts as a “one-shot” pulse generator. The pulse begins when the LM555 timer receives a signal at the trigger input that falls below a 1/3 of the voltage supply. The width of the output pulse is determined by the time constant of an RC network. The output pulse ends when the voltage on the

## Feature Description (continued)

capacitor equals  $2/3$  of the supply voltage. The output pulse width can be extended or shortened depending on the application by adjusting the R and C values.

- Astable (free-running) mode: The LM555 timer can operate as an oscillator and puts out a continuous stream of rectangular pulses having a specified frequency. The frequency of the pulse stream depends on the values of  $R_A$ ,  $R_B$ , and C.

## 7.4 Device Functional Modes

### 7.4.1 Monostable Operation

In this mode of operation, the timer functions as a one-shot (Figure 11). The external capacitor is initially held discharged by a transistor inside the timer. Upon application of a negative trigger pulse of less than  $1/3 V_{CC}$  to pin 2, the flip-flop is set which both releases the short circuit across the capacitor and drives the output high.

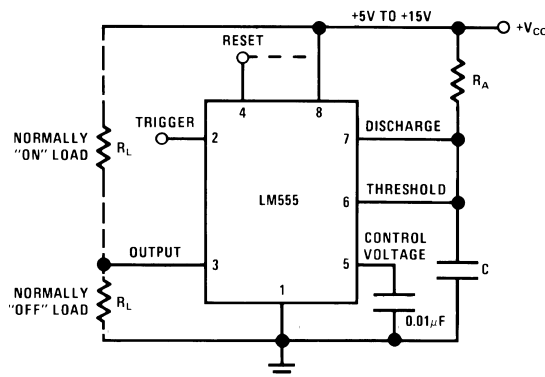
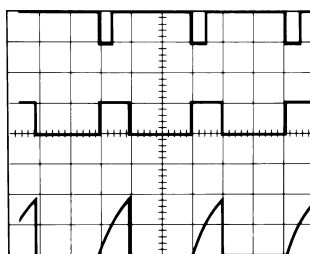


Figure 11. Monostable

The voltage across the capacitor then increases exponentially for a period of  $t = 1.1 R_A C$ , at the end of which time the voltage equals  $2/3 V_{CC}$ . The comparator then resets the flip-flop which in turn discharges the capacitor and drives the output to its low state. Figure 12 shows the waveforms generated in this mode of operation. Since the charge and the threshold level of the comparator are both directly proportional to supply voltage, the timing interval is independent of supply.



$V_{CC} = 5\text{ V}$   
TIME = 0.1 ms/DIV.  
 $R_A = 9.1\text{ k}\Omega$   
 $C = 0.01\text{ }\mu\text{F}$

Top Trace: Input 5V/Div.  
Middle Trace: Output 5V/Div.  
Bottom Trace: Capacitor Voltage 2V/Div.

Figure 12. Monostable Waveforms

During the timing cycle when the output is high, the further application of a trigger pulse will not effect the circuit so long as the trigger input is returned high at least  $10\text{ }\mu\text{s}$  before the end of the timing interval. However the circuit can be reset during this time by the application of a negative pulse to the reset terminal (pin 4). The output will then remain in the low state until a trigger pulse is again applied.

When the reset function is not in use, TI recommends connecting the Reset pin to  $V_{CC}$  to avoid any possibility of false triggering.